BIS FINDER

Accelerating MSE Compliance — Automating BIS Standard Discovery for Building Materials

441	25+	11	RAG
BIS Standards	Categories	Indian Languages + Voice accessibility	AI-Powered Engine

SIGMA SQUAD

IIT Tirupati Hackathon · Bureau of Indian Standards

TEAM

Binary Brains · Vibhaa Selladurai

01 — Problem Statement

Weeks of Manual Research

MSEs spend 2–4 weeks manually searching BIS portals to identify which standards apply — blocking go-to-market timelines.

441 Standards, No Easy Map

BIS covers 441 building material standards across 25+ categories. No structured tool maps product descriptions to the right standards.

Language Barrier

Most MSE owners operate in regional languages. All BIS documentation is in English, creating a deep knowledge access gap.

High Compliance Risk

Wrong or missed standards lead to failed BIS certification, product recalls, and penalties under the BIS Act, 2016.

02 — Solution Overview

Describe a building material product in any Indian language → Get top 3–5 BIS standards with AI rationale in seconds.

RAG Pipeline

- FAISS vector search over 441 BIS standards
- all-MiniLM-L6-v2 → 384-dim embeddings
- Llama 3.1 8B rationale via Databricks API
- Returns top 3–5 standards + compliance rationale

11 Indian Languages

- Hindi, Tamil, Telugu, Kannada, Malayalam
- Bengali, Marathi, Gujarati, Punjabi, Odia, English
- Bidirectional translation via Sarvam mayura:v1
- Full language parity — input and output

Full Voice Support

- Voice input: Sarvam saarika:v2.5 ASR
- Voice output: Sarvam bulbul:v2 TTS
- WAV chunk-based audio delivery
- Accessible to non-literate users

03 — System Architecture

Voice Input
saarika:v2.5
Sarvam ASR

→

Translate IN
mayura:v1
Any lang → EN

→

Embed + Search
MiniLM-L6-v2
FAISS IndexFlatIP

→

LLM Rationale
Llama 3.1 8B
Databricks API

→

Translate + TTS
mayura:v1 + bulbul:v2
WAV output

▶ **User Input (text or voice, any Indian language)**

BIS Standards + Rationale in user's language ◀

Component	Technology
Frontend	Streamlit
Vector Store	FAISS IndexFlatIP (cosine similarity)
LLM	Databricks / Llama-3.1-8B-Instruct
Translation & ASR/TTS	Sarvam AI (mayura:v1, saarika:v2.5, bulbul:v2)

04 — Chunking & Retrieval Strategy

Data Ingestion & Chunking

- Dataset: 441 BIS standards, 25+ building material categories
- Each standard = one document chunk (title + scope + key clauses)
- Chunking strategy: document-level — preserves standard integrity and context
- Encoded with all-MiniLM-L6-v2 → 384-dimensional dense vectors
- L2-normalised for cosine similarity via FAISS IndexFlatIP
- Index stored in-memory at runtime; no persistent vector DB required

Retrieval & Generation Pipeline

1. User query (any language) translated to English via Sarvam mayura:v1
2. Query embedded → 384-dim vector using same MiniLM model
3. FAISS IndexFlatIP cosine search → top-k=5 nearest BIS standards
4. Retrieved standards + query sent to Llama 3.1 8B on Databricks
5. LLM generates rationale + compliance requirements in English
6. Output translated back to user's selected Indian language

05 — Demo Highlights

Text Query

1. Enter Databricks + Sarvam API keys in sidebar
2. Select output language (e.g. Tamil)
3. Type: 'OPC cement for structural use'
4. Receive top 3–5 BIS standards with rationale in Tamil

Voice Query

1. Click microphone → speak in Hindi
2. ASR (saarika:v2.5) transcribes audio to text
3. Pipeline retrieves relevant BIS standards
4. Results read aloud in Hindi via bulbul:v2 TTS

Multilingual Query

1. Input: 'கட்டட கம்பிகள்' (steel rebar) in Tamil
2. Translated to English → embedded → FAISS search
3. Llama 3.1 8B generates rationale in English
4. Response translated back to Tamil for the user

06 — Evaluation Results

87%
Top-3 Hit Rate
on held-out queries

94%
Top-5 Hit Rate
on held-out queries

< 3s
Avg. Response Time
end-to-end text query

441
Standards Indexed
across 25+ categories

Evaluation Metric	Method	Result
Top-3 Standard Match	Manual annotation on 50 test queries	87% correct top-3
Top-5 Standard Match	Manual annotation on 50 test queries	94% correct top-5
Multilingual Fidelity	Back-translation check (EN → lang → EN)	High semantic preservation
Voice Round-trip	ASR → RAG → TTS full pipeline	Functional across 5 languages
LLM Rationale Quality	Relevance + accuracy human eval	Coherent & regulation-aligned

07 — Impact on MSEs

"An MSE owner can now find the right BIS standard for their cement product in seconds — in their own language, by voice."

Time Saved

Reduces compliance discovery from weeks to seconds — freeing MSEs to focus on production.

Reduced Risk

Accurate standard identification minimises failed BIS certifications and costly product recalls.

Language Equity

11 regional languages + voice input makes BIS compliance accessible to every MSE — literate or not.

Broader Reach

Particularly impactful in Tier-2/3 cities and rural construction supply chains.

BIS Act 2016

Helps MSEs comply with mandatory BIS certification under the BIS Act, avoiding penalties.

Faster GTM

Faster standard discovery accelerates product launch timelines for building material manufacturers.

08 — Team & Acknowledgements

My Team

Team Binary Brains
Vibhaa Selladurai
IIT Tirupati Hackathon · Sigma Squad

Built With

- Databricks — Llama 3.1 8B via Foundation Model API
- Sarvam AI — mayura:v1, saarika:v2.5, bulbul:v2
- FAISS — IndexFlatIP vector similarity search
- Hugging Face — all-MiniLM-L6-v2 embeddings
- Streamlit — Frontend + Cloud deployment

Dataset: Official BIS Building Materials standards dataset provided by the hackathon organisers · 441 standards · 25+ categories

Live Demo · Streamlit Cloud
https://hackathongit-5sbttvasahsfw3nkthdf2y.streamlit.app/

GitHub Repository · BIS Finder
https://github.com/Vibhaa2005/hackathon.git